ICH
harmonisation for better health

ICH M15: General Principles for Model-Informed Drug Development

Step 2 document – to be released for comments

29 November 2024

International Council for Harmonisation of Technical Requirements
for Pharmaceuticals for Human Use

Legal Notice

- This presentation is protected by copyright and may, with the exception of the ICH logo, be used, reproduced, incorporated into other works, adapted, modified, translated or distributed under a public license provided that ICH's copyright in the presentation is acknowledged at all times. In case of any adaption, modification or translation of the presentation, reasonable steps must be taken to clearly label, demarcate or otherwise identify that changes were made to or based on the original presentation. Any impression that the adaption, modification or translation of the original presentation is endorsed or sponsored by the ICH must be avoided.
- The presentation is provided "as is" without warranty of any kind. In no event shall the ICH or the authors of the original presentation be liable for any claim, damages or other liability arising from the use of the presentation.
- The above-mentioned permissions do not apply to content supplied by third parties. Therefore, for documents where the copyright vests in a third party, permission for reproduction must be obtained from this copyright holder.

Background

- **This document has been signed off as a *Step 2* document (06 November 2024) to be issued by the ICH Regulatory Members for public consultation**
- **This document was developed based on a Concept Paper (10 November 2022) and a Business Plan (10 November 2022)**
- **Anticipating finalization as a *Step 4* document to be implemented in the local regional regulatory system: 2026**

Key Principles

- For the purposes of this guideline, Model-Informed Drug Development (MIDD) is defined as the strategic use of computational modeling and simulation (M&S) methods that integrate nonclinical and clinical data, prior information, and knowledge (e.g., drug and disease characteristics) to generate evidence.
- This guideline provides general recommendations for planning, model evaluation, and documentation of evidence derived from MIDD, hereafter "MIDD evidence."
- MIDD evidence is defined as model outcomes that have been determined by application of the MIDD evidence assessment framework including Model Evaluation to be appropriate to inform the answer to the Question of Interest.
- This guideline establishes a harmonized assessment framework for MIDD evidence and provides an assessment table as a tool for communication within and between drug developers and regulatory authorities across multidisciplinary teams to increase transparency and provide an understanding of MIDD.

Guideline Objectives and Scope

Objectives

- **To provide general recommendations for planning, Model Evaluation, and documentation of MIDD evidence**
- **To establish a harmonized assessment framework (including associated terminology) for MIDD evidence**
- **To facilitate a multidisciplinary understanding of MIDD and associated evidence generation**

Guideline Objectives and Scope

Scope

- **Applies to both current and emerging M&S methods and applications**
- **Focuses on assessment of MIDD evidence and provides recommendations for related regulatory interactions, reporting, and submission**
- **Does not include details regarding technical aspects of model development. Model development should follow the general recommendations outlined in this guideline.**

Table of Contents

1. INTRODUCTION 4

1.1. Objective of the Guideline 4

1.2. Background 4

1.3. Scope of the Guideline 5

1.4. Outline of the Guideline 5

2. FRAMEWORK FOR ASSESSMENT OF MIDD EVIDENCE 7

2.1. Key Assessment Elements 7

2.2. Additional Considerations for Interaction with Regulators and to Inform Decision-Making 9

3. MODEL EVALUATION 10

4. MIDD REPORTING AND SUBMISSION 13

4.1. Model Analysis Planning (MAP) 13

4.2. Model Analysis Reporting (MAR) 13

4.3. Documentation for Regulatory Interactions and Submissions 13

Appendix 1 TABLE FOR ASSESSMENT OF MIDD EVIDENCE 15

Appendix 2 MODEL ANALYSIS REPORT CONTENT 17

Appendix 3 GLOSSARY 18

Background

- **M&S methods include but are not limited to the following.**
  - Population pharmacokinetics
  - Physiologically based pharmacokinetics and biopharmaceutics
  - Dose-exposure-response
  - Model-based meta-analysis
  - Quantitative systems pharmacology and toxicology
  - Agent-based models
  - Disease progression models
  - Artificial intelligence/machine learning

Outline of the Guideline

Sequence of MIDD in Relation to the Relevant Guideline Sections

Stages	Planning and Regulatory Interaction		Implementation, Reporting, and Submission		
Sequence of Activities	**Key Assessment Elements**	**Additional Considerations for Interaction with Regulator and to Inform Decision-Making**	**Model Evaluation**	**Model Analysis Reporting**	**Documentation for Regulatory Interactions and Submissions**
	• Question of Interest • Context of Use • Model Influence • Consequence of Wrong Decision • Model Risk • Model Impact	• Appropriateness of Proposed MIDD • Technical Criteria for model evaluation and model outcomes[1] These should be documented (e.g., in a Model Analysis Plan [MAP]).	• Verification • Validation • Applicability assessment	• Model Analysis Report(s) (MAR)	• Regulatory documents, including + Outcome of MIDD Evidence Assessment + References to all relevant MAPs and MARs
Relevant Guideline Section	Section 2.1 and Appendix 1	Sections 2.2 and 4.1 and Appendix 1	Section 3	Section 4.2 and Appendix 2	Sections 2 and 4.3 and Appendix 1

Note: Terms used in this table are defined in relevant guideline sections.
[1] Results derived from M&S (i.e., via model-based predictions or simulations) and associated conclusions that are typically aligned to a Question of Interest.

Inform Decision-Making

Framework for Assessment of MIDD Evidence

- **Describes key concepts for assessment of MIDD evidence to inform decision-making, increase transparency, and provide an understanding of MIDD at the planning stage**
- **These concepts are split into:**
  - Key Assessment Elements
  - Additional Considerations for Interactions with Regulators to Inform Decision-Making

Key Assessment Elements

- **Question of Interest:** The question that MIDD is intended to answer
- **Context of Use:** Description of the model(s) and its specific role and scope to answer the Question of Interest
- **Model Influence:** The intended weight of the model outcomes in decision-making considering the contribution of other relevant information
- **Consequence of Wrong Decision:** The consequences (e.g., with respect to patient safety and/or efficacy) if a wrong decision is made, based on all available information
- **Model Risk:** The contribution of the model outcomes to a possible wrong decision and subsequent potential undesirable consequences
- **Model Impact:** The contribution of the model outcomes in relation to current regulatory expectations or standards in answering the Question of Interest

See guideline draft for additional details, including how to derive these elements. For ease of use, see separate Word version of Table for Assessment of MIDD Evidence.

Additional Considerations for Interaction with Regulators

- **MIDD Planning Stage**
  - Appropriateness of Proposed MIDD: Rationale for why the proposed MIDD is suitable to answer the Question of Interest
  - Technical Criteria: Summary and rationale of the key criteria for Model Evaluation and model outcomes to establish acceptability of the model
- **MIDD Evidence Submission Stage**
  - Model Evaluation: Key results and conclusions of the technical evaluation of the model
  - Outcome of the MIDD Evidence Assessment: Multidisciplinary assessment of the MIDD evidence of answer the Question of Interest

See guideline draft for additional details, including guidance with respect to regulatory interactions and submissions. For ease of use, see separate Word version of Table for Assessment of MIDD Evidence.

ICH harmonisation for better health

Table for Assessment of MIDD Evidence

Item	Definition	Instruction	Entry
Key Assessment Elements			
Question of Interest[1]	The question that MIDD is intended to answer.	State the Question of Interest.	
Context of Use	A description of the model(s) and its specific role and scope to answer the Question of Interest.	Provide a concise, clear, and explicit description of the model, the data used to build the model, the specific role of the model outcomes, and the other data or evidence that will contribute to the answer to the Question of Interest.	
Model Influence	The intended weight of the model outcomes in decision-making considering the contribution of other relevant information.	Describe the Model Influence; rate it as low, medium, or high considering other relevant information (e.g., nonclinical and clinical) to inform decision-making; and justify the rating.	
Consequence of Wrong Decision	The consequences (e.g., with respect to patient safety and/or efficacy) if a wrong decision is made, based on all available information.	Describe the consequence of a wrong decision; rate it as low, medium, or high based on the severity of the consequences a wrong decision may have on patient safety and efficacy; and justify the rating.	
Model Risk[2]	The contribution of the model outcomes to a possible wrong decision and subsequent potential undesirable consequences.	Describe the risk; rate it as low, medium, or high based on the Model Influence rate and the Consequence of a Wrong Decision rate; and justify the rating.	
Model Impact	The contribution of the model outcomes in relation to current regulatory expectations or standards in answering the Question of Interest.	Describe the impact; rate it as low, medium, or high considering current regulatory expectations or standards; and justify the rating.	

[1] If MIDD is planned to answer different Questions of Interest, it is recommended to use separate tables for each question.
[2] Model Risk should be interpreted in the context of answering a specific Question of Interest and is not to be perceived as a risk intrinsic to MIDD or M&S.

ICH harmonisation for better health

Table for Assessment of MIDD Evidence

Item	Definition	Instruction	Entry
MIDD Planning Stage[3]			
Appropriateness of Proposed MIDD	The rationale for why the proposed MIDD is suitable to answer the Question of Interest and cover the related key assumptions and required data.	Include a description and justification sufficient to facilitate regulatory interaction on the appropriateness of the proposed MIDD to answer the Question of Interest.	
Technical Criteria	A summary and rationale of the key criteria for Model Evaluation and model outcomes to establish the acceptability of the model (e.g., using an acceptance standard such as bioequivalence acceptance limits).	Include a description of the Technical Criteria for the assessment of Model Evaluation and model outcome. This should include sufficient details on the relevant metric(s).	
MIDD Evidence Submission Stage			
Model Evaluation	A brief discussion of the key results and conclusions of the technical evaluation[4] of the model.	Describe the key results and how they compare to and fulfill the Technical Criteria and conclude on the acceptability of the model performance and model outcome, with details being provided in the appropriate regulatory documentation.	
Outcome of the MIDD Evidence Assessment[5]	A concise summary of the multidisciplinary assessment of the MIDD evidence to answer the Question of Interest.	Provide a multidisciplinary integrative assessment and conclusion for the acceptability of the MIDD evidence to contribute to the answer to the Question of Interest, referring to the MIDD assessment framework elements.	

[3] These items should also be provided at the MIDD Evidence Submission Stage.
[4] Using the principles of Model Evaluation described in Section 3 of the draft guideline, with specific focus on Technical Criteria.
[5] "Assessment" in this context does not refer to any regulatory review activities or processes.

Model Evaluation

- **Provides an overview of Model Evaluation elements and related general recommendations for determination of the acceptability of the model(s) to answer the Question of Interest**
- **Adopting these recommendations ensures that appropriate actions have been taken to inform decision-making.**
- **The Model Evaluation should be commensurate with the Model Risk and be strengthened as it increases.**

See guideline draft for additional details for the general Model Evaluation recommendations.

Model Evaluation

- **Elements**
  - **Verification**: Ensuring user-generated codes are error-free, equations reflecting the model assumptions and their representation in the programming language or software are correct, and calculations are accurate
  - **Validation**: Assessing the adequacy of the model robustness and performance. Validation activities include assessing the relevance and appropriateness of the following: the data, the model's conceptual form (i.e., overall structure and complexity), the model assumptions, the approach to model development, and the graphical and numerical approaches to model performance and external validation.
  - **Applicability Assessment**: Characterizing the relevance and adequacy of the data and model's contribution to answering the Question of Interest

MIDD Reporting and Submission

- **Provides recommendations on MAPs, MARs, and documentation (including the assessment table) with respect to regulatory interactions and submissions**
- **Model Analysis Plan (MAP)**
  - Pre-define and document each model analysis
  - Provision of MAPs during regulatory interactions can facilitate discussions.
- **Model Analysis Report (MAR)**
  - Document results of each model analysis submitted to regulators
  - Key model outcomes described in a single MAR or multiple MARs that support the answer to a Question of Interest should be summarized using the respective assessment table.

See guideline draft for additional details and an appendix outlining MAR content.

MIDD Reporting and Submission

- **Documentation for Regulatory Interactions and Submissions**
  - When MIDD evidence from multiple MARs and/or other sources supports a Question of Interest, an integrated summary should be provided in a concise manner within the assessment table.
  - Additional details should be provided in meeting background materials or Common Technical Document sections, with cross-references to source documentation.
  - The assessment table and all relevant documents (e.g., MAPs, MARs) should be referenced or included in the most appropriate section(s) of the respective regulatory documentation.
  - Inclusion of a summary of relevant regulatory feedback on MIDD is encouraged within background meeting materials and other relevant regulatory documents.

General Considerations

- **This guideline should be used in conjunction with relevant topic-specific ICH guidelines (e.g., E4, E5, E6, E7, S7B, E11[R1]/E11A, E14, M12, E17, and E9/E9[R1]).**

Conclusions

- **Drug development is a sequential and iterative process where MIDD can play an important strategic role.**
- **The guidance on assessment of MIDD evidence, Model Evaluation, and MIDD reporting and submission provided in this guideline is intended to facilitate the use of MIDD evidence to inform decision-making.**
- **The assessment table provides a tool for communication within and between drug developers and regulatory authorities across multidisciplinary teams to increase transparency and enable early alignment to facilitate subsequent acceptance of MIDD evidence.**
- **Additional training materials will be developed to complement this guideline.**

Contact

- **Stakeholders from ICH Member countries/regions are encouraged to submit their comments to their respective Regulatory Authorities or, if not covered by an ICH regulatory Authority, directly to ICH as per instructions at this link.**
- **For any questions, please contact the ICH Secretariat:**

admin@ich.org